CYNARIS SOLUTIONS — MODEL CARD

MODEL CARD: ADULT INCOME CLASSIFICATION

1. MODEL DETAILS
Model: Logistic Regression
Dataset: UCI Adult Income Dataset
Framework: Scikit-learn and IBM AI Fairness 360
Protected Attribute: Gender

2. INTENDED USE
Experimental binary classification of income categories
using demographic and employment-related features.
This model is intended for educational fairness analysis only.

3. MODEL LIMITATIONS
The dataset represents a particular historical population.
Its predictions should not be treated as objective measures
of individual ability, worth, or financial reliability.

4. EVALUATION DATA
The dataset was divided into 70% training and 30% testing data.
The same test dataset was used for both models.

5. FAIRNESS EVALUATION

Baseline Model:
Disparate Impact: 0.3430
Statistical Parity Difference: -0.1703
Equal Opportunity Difference: -0.0660

After Reweighing:
Disparate Impact: 0.6127
Statistical Parity Difference: -0.0833
Equal Opportunity Difference: 0.1594

6. MITIGATION
AIF360 Reweighing was applied to training instances.
The generated instance weights were used during
Logistic Regression model training.

7. EXPLAINABILITY
SHAP LinearExplainer was used to identify the contribution
of input features to model predictions.

8. ETHICAL CONSIDERATIONS
Gender-based disparities may remain even when gender is
excluded from predictive features because other variables
can act as proxies. Fairness metrics do not prove that a
model is completely unbiased.

9. CAVEATS
This is an experimental educational model.
It has not been validated for real-world employment,
credit, lending, or other high-impact decisions.